

Appointment Booking System – Book Time Slots

Project Documentation

APPOINTMENT BOOKING SYSTEM

Mini Project Documentation

Project Title: Appointment Booking System – Book Time Slots

Submitted By: Shaik Reshma, Shaik Moien, N. Vishnu Vardhan

Course: B.Tech – Computer Science and Engineering

Programming Language: Java

Project Type: Console-Based Application

Abstract

The Appointment Booking System is a console-based Java application developed to simplify and automate the process of booking appointments. In many organizations such as hospitals, clinics, salons, banks, educational institutions, and other service centers, managing appointments manually is time-consuming and often results in scheduling conflicts, duplicate bookings, and inefficient record management. This project provides a simple and user-friendly solution by allowing users to manage appointments through a menu-driven console interface.

The application enables users to view available time slots, book appointments, cancel existing bookings, and view all booked appointments. The system maintains appointment details in an organized manner and ensures that a booked time slot cannot be assigned to another user until it is canceled. This helps reduce errors, improves time management, and enhances the overall efficiency of the appointment scheduling process.

The project is developed using the Java programming language and demonstrates the practical implementation of Object-Oriented Programming (OOP) concepts such as classes, objects, methods, encapsulation, loops, conditional statements, and collections like ArrayList. The application also uses the Scanner class to accept user input and provides a simple menu-driven interface that is easy to understand and operate.

This project is designed mainly for educational purposes and helps students gain practical knowledge of Java programming by developing a real-world application. It improves problem-solving skills and provides experience in designing and implementing console-based software systems. The Appointment Booking System can be further enhanced by integrating a graphical user interface (GUI), database connectivity, online appointment scheduling, user authentication, email notifications, and report generation. Overall, this project serves as an effective learning tool while providing a simple, reliable, and efficient appointment management solution.

1. Introduction

Appointment booking is an important process in hospitals, salons, banks, educational institutions, and many service organizations. Manual booking systems consume more time and may lead to scheduling conflicts.

This project automates the booking process by allowing users to choose available time slots and manage appointments through a simple console interface.

2. Objectives

- Develop a console-based appointment booking application.
- Allow users to view available slots.
- Book appointments.
- Cancel booked appointments.
- Display booked appointments.
- Learn Java Object-Oriented Programming concepts.

3. Problem Statement

Manual appointment booking is time-consuming and may result in duplicate bookings or scheduling errors. A computerized appointment booking system helps manage appointments efficiently and reduces human errors.

4. Scope of the Project

- Salons
- Tuition centers
- College laboratories
- Personal appointment management

5. System Requirements

Hardware Requirements:

- Processor: Intel Core i3 or above

- RAM: 4 GB
- Hard Disk: 100 MB Free Space

Software Requirements:

- Operating System: Windows/Linux/Mac
- Java JDK 17 or above
- VS Code / IntelliJ IDEA / Eclipse
- Command Prompt or Terminal

6. Modules

Module 1 – Login

Allows the user to log in using a username and password.

Module 2 – View Available Time Slots

Displays all available appointment slots.

Module 3 – Book Appointment

Allows users to select and book an available slot.

Module 4 – Cancel Appointment

Cancels a previously booked appointment.

Module 5 – View Booked Appointments

Displays all booked appointments.

Module 6 – Exit

Closes the application.

7. Functional Requirements

- User Login
- Display Available Slots
- Book Appointment
- Cancel Appointment
- View Bookings
- Exit Program

8. Non-Functional Requirements

- User Friendly
- Fast Execution
- Reliable
- Simple Interface
- Easy Maintenance

9. Algorithm

Step 1

Start the program.

Step 2

Display the login screen.

Step 3

Verify username and password.

Step 4

Display the main menu.

Step 5

If user selects View Slots, display available slots.

Step 6

If user selects Book Appointment, save the selected slot.

Step 7

If user selects Cancel Appointment, remove the booking.

Step 8

If user selects View Bookings, display booked appointments.

Step 9

If user selects Exit, terminate the program.

10. Flowchart

START

|

Login

|

Login Successful?

/ \

No

Yes

|

|

Login Again

Main Menu

|

|

|

|

|

|

View Book Cancel View Exit

Slots Appointment Bookings

|

|

|

|

-----Return to Menu---

|

Exit

|

END

11. Output Screens

Login

===== **LOGIN** =====

Username : admin

Password : admin123

Login Successful

Main Menu

===== **APPOINTMENT BOOKING SYSTEM** =====

1. View Available Slots
2. Book Appointment
3. Cancel Appointment
4. View Booked Appointments
5. Exit

Enter Choice:

Booking

Available Slots

1. 09:00 AM
2. 10:00 AM
3. 11:00 AM

Enter Slot Number

Appointment Booked Successfully

Cancel Booking

Booked Appointment Cancelled Successfully

View Booking

Booked Appointments

09:00 AM

11:00 AM

12. Advantages

- Easy to use
- Saves time
- Reduces manual work
- Prevents duplicate booking
- Simple and efficient

13. Limitations

- Console-based interface
- Single-user system
- Data is not stored permanently (unless file handling is added)

14. Future Enhancements

- MySQL Database Integration
- Java Swing GUI
- Online Appointment Booking
- Email Notifications
- Admin Dashboard
- Multi-user Support

15. Conclusion

The Appointment Booking System is a simple Java console application that demonstrates the practical use of Java programming concepts. It simplifies appointment management by allowing users to view, book, cancel, and manage appointment slots efficiently. The project is easy to understand and suitable for beginners and B.Tech mini-project submissions.

16. References

1. Java Documentation – Oracle
2. Java Programming by Herbert Schildt
3. Introduction to Java Programming by Y. Daniel Liang
4. Eclipse IDE Documentation